

Time-Series JEPA for Predictive Remote Control Under Capacity-Limited Networks

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Abstract—In remote control systems, transmitting large data volumes (e.g., images and video frames) from wireless sensors to remote controllers is challenging when uplink capacity is limited (e.g., RedCap devices or massive wireless sensor networks). Furthermore, controllers often need only information-rich representations of the original data. To address this, we propose a semantic-driven predictive control combined with a channel-aware scheduling to enhance the control performance for multiple devices under limited network capacity. At its core, the proposed framework, coined time-series joint-embedding predictive architecture (TS-JEPA), encodes high-dimensional sensory data into low-dimensional semantic embeddings at the sensor, reducing communication overhead. Furthermore, TS-JEPA enables predictive inference by predicting future embeddings from current ones and predicted commands, which are directly used by a semantic actor model to compute control commands within the embedding space, eliminating the need to reconstruct raw data. To further enhance reliability and communication efficiency, a channel-aware scheduling is integrated to dynamically prioritize device transmissions based on channel conditions and age of information (AoI). Extensive simulations on inverted cart-pole systems demonstrate that the proposed framework achieves a 98.95% reduction in communication cost, a normalized prediction error of 0.004, and 74.48% control accuracy, outperforming conventional control baselines. Furthermore, the proposed framework maintains robust control performance on 15-step prediction horizons and supports up to $16\times$ more devices than round-robin and opportunistic scheduling schemes with conventional control baselines under similar network conditions.

Index Terms—Joint-embedding predictive architecture, predictive control, self-supervised learning, semantic communication.

NOMENCLATURE

$\mathbf{x}_{i,k}$	State of device i at time k .
p	Dimensions of device state.
$\mathbf{u}_{i,k}$	Control command of device i at time k .
τ_o	Fixed sampling time.
q	Dimensions of control commands.
$\mathbf{n}_{s,k}$	Process noise vector at time k .

N_s	Gaussian process noise variance.
$\mathbf{f}_i$	Nonlinear dynamics of device i .
$\mathcal{J}(\cdot, \cdot)$	Quadratic cost function.
$\mathbf{R}$	Control effort positive definite matrix.
x_d	Desired cart position.
$\text{PL}_{\text{LoS}}^{\text{dB}}$	Line-of-sight path-loss in dB.
$D_i^{3\text{-D}}$	3-D distance between device i and remote controller.
W_c	Center frequency.
$\mathbb{P}_{\text{LoS}}$	Channel probability in line-of-sight.
$D_i^{2\text{-D}}$	2-D distance between device i and remote controller.
D_{clutter}	Clutter size.
δ	Clutter density.
h_c	Clutter height.
$h_{i,R}$	Antenna height of the i th device.
h_{BS}	BS's antenna height.
$\text{PL}_{\text{LoS}}^{\text{dB}}$	Non-line-of-sight path loss in dB.
P_i	Transmission power of the i th device.
P_{max}	Maximum transmit power allowed per device.
$\gamma_{i,k}$	Signal-to-noise ratio (SNR) of the device i at time k .
$H_{i,k}$	Channel gain between the i th device and base station at time k .
N_c	Additive white Gaussian noise power.
W_i	Allocated transmission bandwidth for the device i .
$R_{i,k}$	Uplink channel capacity for the device i at time k .
$\epsilon_{i,k}$	Likelihood of unsuccessful transmission.
$\Psi_{\theta}(\cdot)$	Context encoder parametrized by a set of parameters θ .
$\Psi_{\bar{\theta}}(\cdot)$	Target encoder parametrized by a set of parameters $\bar{\theta}$.
$\mathbf{z}_{i,k}$	Low-dimensional embeddings of the device i at time k .
$\mathcal{P}_{\varphi}(\cdot)$	Predictor parametrized by a set of parameters φ .
$\hat{\mathbf{u}}_{i,k}$	Predicted control-command of device i at time k .
K_p	Prediction horizon.
$\tilde{\mathbf{z}}_{i,k}$	Predicted embeddings of device i at time k .
η	Decay rate regulates the weight update rate.

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$\mathcal{D}_s$	Time-series joint-embedding predictive architecture (TS-JEPA) dataset.
K_s	Number of consecutive states and their commands.
$\mathcal{D}_a$	Semantic actor dataset.
K_a	Number of consecutive embeddings and their commands.
$\alpha_{i,k}$	Scheduling decision of the device i at time k .
$\beta_{i,k}$	Age of information (AoI) of the device i at time k .
γ_{th}	SNR threshold.
$\beta_{i,th}$	AoI threshold.
J	Number of available resource blocks.
I	Total number of devices.
$Q_{i,k}$	Virtual queue of device i at time k .
S_k	Network state at time k .
$\mathbf{Q}_k$	Virtual queues vector at time k .
$\mathcal{L}(\cdot)$	Lyapunov function.
$\Delta(\mathbf{Q}_k)$	Expected change in Lyapunov function.
$P_{i,k}^{req}$	Minimum required transmission power.
$S_{i,k}$	Drift-plus-penalty index of device i at time k .
$\mathcal{N}_{i,k}^P$	Mean absolute percentage error of device i at time k .
$\mathcal{N}_{i,K_p}^u$	Normalized control prediction error of device i
$\mathcal{R}_{i,k}$	Scoring function of device i at time k .
κ	Number of consecutive frames.
$x_{i,k}$	Cart location of device i at time k .
$\theta_{i,k}$	Pendulum angle of device i at time k .

I. INTRODUCTION

SEMANtic communication has emerged as a critical enabler for the efficiency and scalability of next-generation 6G applications [1], [2], such as autonomous systems, smart cities, and immersive augmented reality [3], [4], [5]. In contrast to 5G technologies, where ultrareliable low-latency communication (URLLC) focuses on achieving high reliability and ultralow latency at the cost of significant communication resources [6], and massive machine-type communication (mMTC) prioritizes the massive connectivity with reduced quality-of-service [7], semantic communication shifts the emphasis toward extracting and transmitting only the most relevant and meaningful features of the data. This paradigm aims to enhance both communication and control efficiency by reducing redundant transmissions and focusing on extracting and leveraging the intrinsic meaning embedded within the data.

A key aspect of semantic communication in control applications is the ability to represent high-dimensional sensor data (e.g., images or video frames) as low-dimensional semantic embeddings. These embeddings maintain the information necessary for downstream tasks while discarding redundancies and reducing the communication overhead. As a result, efficient resource management for transmitting semantic embeddings becomes vital, particularly in limited wireless networks, to ensure timely and accurate control decisions for

large-scale control systems. This semantic-driven approach not only improves network scalability and reduces latency but also aligns closely with the growing need for intelligence at the edge, where devices transmit only distilled semantic information rather than raw data. This shift in communication priorities motivates the integration of semantic representations with dynamic scheduling and machine learning (ML) techniques, as discussed in Sections III and IV.

A. Related Work

Efficient scheduling is a fundamental requirement for scalable and reliable remote control in next-generation 6G wireless networks. Existing scheduling strategies, including round-robin [8], [9], opportunistic [10], [11], [12], stability-aware [13], and control-aware approaches [14], have been developed to allocate wireless resources while maintaining the acceptable control performance. However, these approaches typically rely on high-dimensional data transmissions and are not designed to optimize the semantic efficiency of transmitted information. Moreover, their scalability is often limited by the increasing computational and communication overhead as the number of devices grows.

Recent advances in ML offer promising alternatives by enabling predictive modeling of device dynamics and encoding raw sensory data into task-relevant representations. In particular, existing approaches in this domain can be broadly categorized into joint-embedding architecture (JEA) and generative architectures. The JEAs aim to learn representations such that embeddings of semantically similar inputs are mapped close together, while dissimilar ones are mapped apart. Popular examples include simple framework for contrastive learning of visual representations (SimCLR) [15] and bootstrap your own latent (BYOL) [16]. Although the JEAs are computationally efficient and avoid pixel-level reconstruction, they are prone to representation collapse, where all inputs converge to a single embedding. To mitigate this, strategies such as contrastive losses and momentum encoders have been proposed [17], [18].

On the other hand, generative architectures, including recurrent neural networks (RNNs)-based models [19], [20], long short-term memory (LSTM) [21], gated recurrent unit (GRU) [22], convolution neural networks (CNNs) [23], and variational autoencoder (VAE) [24], reconstruct high-dimensional data from conditional inputs. These generative models are robust to representation collapse but often incur high computational cost and require large amounts of labeled data, making them less suitable for real-time remote control with limited wireless resources.

The joint-embedding predictive architecture (JEPA) [25], [26] offers a middle ground by predicting only relevant aspects of target data in a latent space, rather than reconstructing it in a raw high-dimensional space. Specifically, the JEPAs seek to predict the embeddings of target high-dimensional data from the embeddings of input data and a conditional context, with a loss computed in the latent space, making it computationally efficient and scalable. Applications of JEPAs have shown success in various domains, including image [26],

TABLE I
COMPARISON BETWEEN TS-JEPA AND EXISTING LEARNING APPROACHES

Aspect	TS-JEPA (Proposed)	I-JEPA [26]	V-JEPA [27]	Generative Architectures [24]	Supervised Learning
Primary Objective	Learn action-conditioned latent dynamics by predicting future embeddings of control systems.	Learn semantic representations for static images.	Learn spatio-temporal representations for general videos.	Reconstruct high-dimensional data from latent representations.	Minimize task-specific prediction loss using labeled data.
Learning Paradigm	Self-supervised predictive joint-embedding with control loss.	Self-supervised joint-embedding.	Self-supervised temporal joint-embedding.	Generative with pixel-level reconstruction.	Supervised learning using labeled data.
Input Modality	Control-system frames and predicted control commands.	Static images.	Video frames (temporal context).	Images, time-series, or signals.	Task-specific labeled inputs.
Prediction Target	Future semantic embeddings (latent prediction).	Masked image embeddings.	Masked or future video embeddings.	Full pixel-space reconstruction.	Output labels or control targets.
Loss Computation	Latent-space, control-aligned loss (cosine similarity).	Latent-space similarity.	Latent-space similarity.	Pixel-space reconstruction error (MSE/ELBO).	Task-specific loss (MSE/cross-entropy).
Unique Contribution	Link semantic representation learning with non-linear control.	Learns efficient image representations.	Learns general video world models.	Models data generation, not semantics.	Direct input-output mapping, lacks interpretability.

video [27], and audio [28], demonstrating their potential for real-time control under limited network capacity.

B. Contributions and Organization

To address the critical challenge of remotely controlling multiple devices that share a limited wireless uplink to transmit high-dimensional sensory data, our key contributions are summarized as follows.

- 1) We propose a novel TS-JEPA that encodes high-dimensional sensory data at the device into low-dimensional semantic embeddings. The TS-JEPA captures the latent dynamics, enabling the prediction of future embeddings at the remote controller without reconstructing high-dimensional sensory data. This significantly reduces communication overhead without compromising the control performance. To further highlight the proposed time-series joint-embedding predictive architecture (TS-JEPA) compared to the existing approaches, including I-JEPA [26], V-JEPA [27], and generative architecture [21], [22], [24], Table I is presented to outline the key similarities and differences in objectives, learning paradigms, and input modality.
- 2) On top of TS-JEPA embeddings, we train a *semantic actor model* that directly maps low-dimensional semantic embeddings to control commands. This avoids the need to reconstruct high-dimensional data, efficiently reducing communication and computation costs.
- 3) We develop a *channel-aware scheduling* to dynamically prioritize devices for transmission based on channel conditions and the AoI, ensuring that the most time-sensitive and reliable updates are transmitted under limited network capacity.
- 4) Extensive simulations on inverted cart-pole systems demonstrate that the proposed framework achieves control performance comparable to conventional baselines while significantly reducing communication overhead. Furthermore, the proposed approach scales effectively to multidevice scenarios, highlighting its practical applicability for remote control under limited network capacity.

The remainder of this article is organized as follows. Section II describes the system model and the problem statement. In Section III, we present the proposed semantic-driven pre-

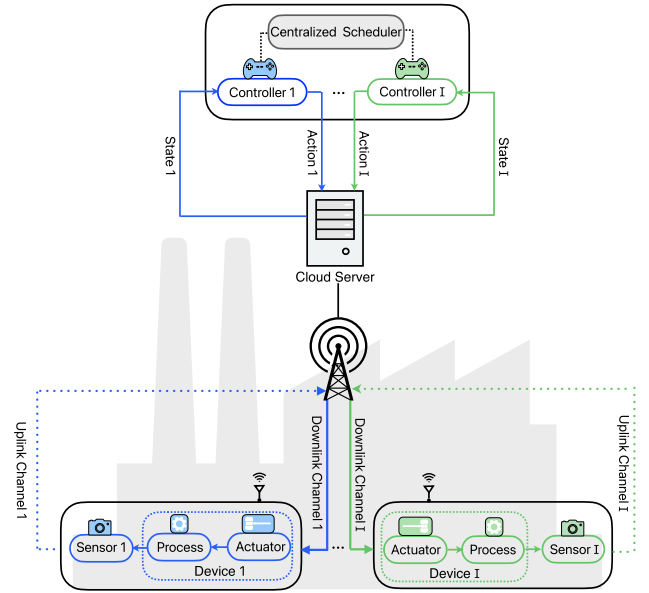


Fig. 1. Illustration of wireless-networked control systems in a smart factory.

dictive control, including TS-JEPA and the semantic actor model, combined with channel-aware scheduling. Section IV presents simulation results that validate the effectiveness of the proposed approach in large-scale wireless control scenarios with limited network capacity. Finally, Section V concludes this article and outlines future work.

II. SYSTEM MODEL

We consider a wireless-networked control system, as shown in Fig. 1, comprising multiple independent nonlinear control systems that operate over shared wireless channels. Each control system includes a device, a sensor, and a remote controller. The device integrates a nonlinear dynamic *process* that is being controlled along with an *actuator* responsible for applying control commands to drive the process toward its desired state. Each device is paired with a *sensor* that periodically samples the device state to monitor its behavior. The sampled state is transmitted via *uplink transmission* to a remote controller endowed with substantially higher com-

putational resources than the device itself, enabling advanced model training and nonlinear control-command computation that are infeasible on a resource-constrained device. Upon receiving the device state, the remote controller computes the appropriate control command and transmits it back to the actuator through downlink transmission to ensure that the process is steered toward its desired state. Further details on control and communication systems are discussed in the following sections.

A. Control System

We consider a wireless-networked control system consisting of multiple independent devices, each equipped with a sensor and an actuator, communicating with a remote controller over wireless channels. Each device is indexed by $i \in \mathcal{I}$ with cardinality $|\mathcal{I}|$. The sensor periodically samples the device's p -dimensional state vector at a fixed sampling time τ_o , where the sampled state for device $i \in \mathcal{I}$ at time $t = k\tau_o$ is represented as $\mathbf{x}_{i,k} \in \mathbb{R}^p$. Upon receiving the sampled state $\mathbf{x}_{i,k}$, the remote controller computes a target q -dimensional control-command vector $\mathbf{u}_{i,k} \in \mathbb{R}^q$.

The calculated control command is then transmitted over ideal channels to the actuator, which applies it to the corresponding device. The state evolution of device i follows the discrete-time nonlinear dynamics given as follows [29]:

$$\mathbf{x}_{i,k+1} = \mathbf{f}_i(\mathbf{x}_{i,k}, \mathbf{u}_{i,k}) + \mathbf{n}_{s,k} \quad (1)$$

where $\mathbf{n}_{s,k} \in \mathbb{R}^p$ is a process noise vector modeled as an independent and identically distributed (i.i.d.) Gaussian random variables with zero mean and variance N_s . The nonlinear function $\mathbf{f}_i : \mathbb{R}^p \times \mathbb{R}^q \rightarrow \mathbb{R}^p$ maps the current state and control command to the next state, representing the nonlinear dynamics of the device i .

To ensure optimal control, the remote controller computes the target control command utilizing the nonlinear control policy to solve the optimization problem formulated as follows [30]:

$$\begin{aligned} \mathbf{u}_{i,k}^* &= \arg \min_{\mathbf{u}_{i,k}} \mathcal{J}(\mathbf{x}_{i,k}, \mathbf{u}_{i,k}) \\ \text{subject to: } &(1) \\ &\mathbf{u}_{\min} \leq \mathbf{u}_{i,k} \leq \mathbf{u}_{\max} \end{aligned} \quad (2)$$

where the quadratic cost function is defined as follows:

$$\mathcal{J}(\mathbf{x}_{i,k}, \mathbf{u}_{i,k}) = \frac{1}{2} \sum_{k=0}^K \|\mathbf{x}_{i,k} - \mathbf{x}_d\|_F^2 + \mathbf{u}_{i,k}^T \mathbf{R} \mathbf{u}_{i,k} \quad (3)$$

where $\|\cdot\|_F$ is the Frobenius norm capturing the state deviation from the desired state $\mathbf{x}_d$, while $\mathbf{R}$ is a positive definite matrix penalizing control effort. Solving the optimization problem in (2) through dynamic programming [31], [32] yields the target control commands. As depicted in Fig. 1, each device's control loop is based on wireless channels for state transmission to ensure successful control performance. Since the device dynamics are inherently unstable, uplink transmission failures prevent the application of appropriate control commands, causing the state $\mathbf{x}_{i,k}$ to diverge to infinity as $k \rightarrow \infty$.

B. Wireless Communication System

We consider a sparse clutter and high base station indoor factory (InF-SH) scenario, where regularly structured devices are randomly distributed within the coverage area of the base station. As shown in Fig. 1, remote controllers are assumed to be located on a distant cloud server with negligible communication delay between the cloud server and the base station. The communication between the base station and the devices occurs through dedicated data and control channels. The data channels facilitate the exchange of device states and control commands, while the control channels manage state information such as scheduling requests and grants.

At each time step $t = k\tau_o$, the centralized scheduler at the base station dynamically manages the uplink channel access by issuing scheduling grants via error-free downlink control channels. These control channels operate without contention or collisions, ensuring reliable command delivery. The data channels follow a standard path loss and Rayleigh block fading model [33], where the channel gains for each device remain constant over the duration of τ_o but vary independently across different time intervals. In the InF-SH scenario, the line-of-sight (LoS) path loss between the i th device and its remote controller is given as follows [34], [35]:

$$\text{PL}_{\text{dB}}^{\text{LoS}} = 31.84 + 21.5 \log_{10}(D_i^{3\text{-D}}) + 19 \log_{10}(W_c) \quad (4)$$

where $D_i^{3\text{-D}}$ denotes the 3-D distance (in meters) between the i th device and the remote controller, while W_c represents the center frequency (in GHz).

The probability of the channel being in an LoS at a given distance is modeled as follows:

$$\mathbb{P}_{\text{LoS}} = \exp \left[-\frac{D_i^{2\text{-D}}}{-\frac{D_{\text{clutter}}}{\ln(1-\delta)}} \cdot \frac{h_{\text{BS}} - h_{i,R}}{h_c - h_{i,R}} \right] \quad (5)$$

where $D_i^{2\text{-D}}$ is the 2-D distance between the i th device and its remote controller. In addition, D_{clutter} , δ , h_c , $h_{i,R}$, and h_{BS} denote the clutter size, clutter density, clutter height, antenna height of the i th device, and the base station's antenna height, respectively. If the channel is classified as non-LoS, the path loss between the i th device and its remote controller is calculated as follows:

$$\text{PL}_{\text{dB}}^{\text{NLoS}} = \max(\text{PL}_{\text{dB}}^{\text{LoS}}) \quad (6)$$

where the general path loss model PL_{dB} in (6) is given by [34], [35]

$$\text{PL}_{\text{dB}} = 33.63 + 21.9 \log_{10}(D_i^{3\text{-D}}) + 20 \log_{10}(W_c) \quad (7)$$

with a shadow fading standard deviation of 4.0.

Here, we consider uplink data transmission from the i th device to its corresponding remote controller at time $t = k\tau_o$ with transmission power $P_i \in [0, p_{\max}]$ of a maximum transmission power $p_{\max}$, the received signal-to-noise ratio (SNR) at the base station is given as follows:

$$\gamma_{i,k} = 10^{-\frac{\text{PL}_{\text{dB}}^{\text{NLoS}}}{10}} \frac{P_i |H_{i,k}|^2}{N_c} \quad (8)$$

where $|H_{i,k}|^2$ denotes the Rayleigh flat-fading channel power gain between the i th device and the base station at time

k , while N_c represents the additive white Gaussian noise (AWGN) power. The uplink channel capacity for device i at time k is given as follows:

$$R_{i,k} = W_i \log_2 (1 + \gamma_{i,k}) \quad (9)$$

where W_i is the allocated transmission bandwidth. Reliable communication is crucial to ensuring effective control performance. However, an outage transmission occurs when the uplink channel capacity falls below a predefined threshold $\bar{R}$, which guarantees the required transmission rate for stable control. The outage probability, representing the likelihood of unsuccessful transmission, is given by

$$\begin{aligned} \epsilon_{i,k} &= \mathbb{P}[R_{i,k} < \bar{R}] \\ &= 1 - \exp \left[-10^{\frac{\text{Pr}_{\text{LoS}}}{10}} \frac{N_c}{P_i} \left(2^{\frac{\bar{R}}{W_i}} - 1 \right) \right] \end{aligned} \quad (10)$$

which follows from the cumulative distribution function (cdf) of the exponential random variable. This outage probability highlights the dependence on key factors such as transmission power P_i , allocated bandwidth W_i , and device-to-controller distance D_i^{2-D} . Hence, efficient utilization of wireless resources is essential to minimize communication failures and ensure reliable uplink transmission, which is critical for maintaining robust control performance in wireless-networked control systems.

C. Problem Statement

In the considered wireless-networked control system, ensuring timely and reliable state updates from devices to remote controllers is vital for maintaining robust control performance. However, the presence of multiple devices that transmit high-dimensional states over limited wireless resources introduces a significant challenge. This challenge introduces a critical tradeoff between communication efficiency and control performance. Hence, effective scheduling and efficient resource allocation are essential to balance these competing objectives for maintaining control performance under limited network capacity. Existing solutions primarily rely on time-triggered or round-robin scheduling, which struggles to adapt to dynamic network conditions and varying control requirements [8]. While recent approaches have explored adaptive transmission based on control states and channel conditions [10], [14], [36], these approaches often lack scalability when managing multiple devices due to their dependence on the transmission of high-dimensional states. Throughout this work, the terms high-dimensional state and frame are used interchangeably to reflect visual observations of the device state captured by the sensor.

To address these limitations, we introduce a novel *semantic-driven predictive control* combined with a *channel-aware scheduling* to enhance the control performance for multiple devices operating under limited network capacity. Unlike traditional approaches that rely on transmitting raw high-dimensional states, our proposed framework efficiently encodes these states into a low-dimensional embedding space, significantly reducing communication overhead without compromising the control performance. The proposed

semantic-driven predictive control learns a mapping $\Psi(\cdot)$ that transforms the high-dimensional device state $\mathbf{x}_{i,k}$ at time $t = k\tau_o$ into a low-dimensional embedding $\mathbf{z}_{i,k} = \Psi(\mathbf{x}_{i,k})$. This embedding space captures the essential latent dynamics, enabling the prediction of future embeddings $\mathbf{z}_{i,k+K_p}$ directly from the current embedding and a sequence of predicted control commands $(\tilde{\mathbf{u}}_{i,k}, \dots, \tilde{\mathbf{u}}_{i,k+K_p-1})$. Specifically, a second mapping $\mathcal{P}(\cdot)$ is introduced to predict future embeddings given as follows:

$$\tilde{\mathbf{z}}_{i,k+K_p} = \mathcal{P}(\mathbf{z}_{i,k} | \tilde{\mathbf{u}}_{i,k}, \dots, \tilde{\mathbf{u}}_{i,k+K_p-1}) \quad (11)$$

where K_p represents the prediction horizon and $\tilde{\mathbf{u}}_{i,k}$ denotes the predicted control command of device i at time $t = k\tau_o$. This predictive capability eliminates the need to reconstruct the raw high-dimensional state for control-command computation. Instead, low-dimensional embeddings are directly leveraged to compute control commands, optimizing for downstream control tasks under limited communication resources. By transmitting low-dimensional embeddings instead of high-dimensional states, the proposed semantic-driven predictive control significantly reduces the wireless communication overhead, allowing more devices to efficiently share limited wireless resources without compromising the control performance. To further enhance reliability and communication efficiency, a channel-aware scheduling is integrated with the semantic-driven predictive control. This scheduling approach dynamically selects devices for transmission based on their channel conditions and age of information (AoI), ensuring that the most critical updates are prioritized for transmission. By balancing channel quality and information freshness, the channel-aware scheduling maximizes resource utilization, minimizes transmission failures, and effectively mitigates the outdated control information at remote controllers.

III. SEMANTIC-DRIVEN PREDICTIVE CONTROL WITH CHANNEL-AWARE SCHEDULING

To address the key challenges in the wireless-networked control system, this section introduces a novel semantic-driven predictive control combined with channel-aware scheduling. The proposed framework leverages a TS-JEPA enhanced with a semantic actor model to improve communication efficiency, reduce latency and maintain robust control performance in large-scale devices under limited wireless resources.

A. Time-Series Joint-Embedding Predictive Architecture

The proposed TS-JEPA is a self-supervised learning technique designed to address the challenges in the wireless network control system by efficiently encoding high-dimensional device states into low-dimensional embeddings while enabling the accurate prediction of future embeddings.

Unlike image-JEPA (I-JEPA) [26] and video-JEPA (V-JEPA) [27], which focus on learning semantic representation for static or visual scenes, TS-JEPA learns action-conditioned latent dynamics of a control system by predicting future semantic embeddings from the current system frame and predicted control commands. In contrast to Dreamer [37], which trains a generative world model

through reinforcement learning and optimizes an energy-free or expected-return principle, TS-JEPA adopts a discriminative joint-embedding objective that directly minimizes a control-aligned loss in latent space. Furthermore, TS-JEPA is explicitly designed for networked control scenarios, where the on-device encoder operates under limited computational and communication resources, and the remote controller, with high computational capacity, performs future-embedding prediction and control-command computation. This design enables scalable, low-overhead, and semantically rich predictive control, distinguishing TS-JEPA from prior JEPA frameworks.

The key components of TS-JEPA, as illustrated in Fig. 2, are described as follows.

- 1) *Context*: At time step $t = k\tau_o$, the context encoder $\Psi_\theta(\cdot)$, parameterized by a set θ of learnable parameters, processes the high-dimensional state $\mathbf{x}_{i,k}$ and maps it to a low-dimensional embedding $\mathbf{z}_{i,k} = \Psi_\theta(\mathbf{x}_{i,k})$.
- 2) *Targets*: For a defined prediction horizon K_p , the target encoder $\Psi_{\bar{\theta}}(\cdot)$ processes a sequence of future device states $(\mathbf{x}_{i,k+1}, \dots, \mathbf{x}_{i,k+K_p})$ to generate their corresponding low-dimensional embeddings $(\mathbf{z}_{i,k+1}, \dots, \mathbf{z}_{i,k+K_p})$.
- 3) *Predictions*: The predictor $\mathcal{P}_\varphi$, parameterized by a set φ of learnable parameters, captures the nonlinear embedding evolution to predict target embedding directly from the current embedding and predicted control commands. Specifically, the predictor infers the future embeddings based on current embeddings and predicted control commands $(\tilde{\mathbf{u}}_{i,k}, \dots, \tilde{\mathbf{u}}_{i,k+K_p-1})$, solving an autoregressive task as follows:

$$(\tilde{\mathbf{z}}_{i,k+1}, \dots, \tilde{\mathbf{z}}_{i,k+K_p}) = \mathcal{P}_\varphi(\mathbf{z}_{i,k} | \tilde{\mathbf{u}}_{i,k}, \dots, \tilde{\mathbf{u}}_{i,k+K_p-1}) \quad (12)$$

where $\tilde{\mathbf{z}}_{i,k}$ is the predicted embeddings at time k .

- 4) *Loss*: To train TS-JEPA, a cosine similarity loss function is employed to align the predicted embeddings with their target embeddings. The context encoder's and predictor's parameters θ and φ are jointly learned through gradient-based optimization as follows:

$$\arg \min_{\theta, \varphi} \frac{1}{K_s} \sum_{k=1}^{K_s} \frac{\langle \tilde{\mathbf{z}}_{i,k+1}, \mathbf{z}_{i,k+1} \rangle}{\|\tilde{\mathbf{z}}_{i,k+1}\|_2 \cdot \|\mathbf{z}_{i,k+1}\|_2} \quad (13)$$

while the target encoder's parameters $\bar{\theta}$ are updated, at each training step, via an exponential moving average (EMA) of the context encoder's parameters to ensure stable training. The target encoder's parameters update rule is given as follows:

$$\bar{\theta} \leftarrow \eta \bar{\theta} + (1 - \eta) \theta \quad (14)$$

where $\eta \in [0, 1]$ is a decay rate that regulates the weight update rate [16].

Specifically, the target encoder mirrors the context encoder's architecture and shares identical parameters at initialization. Since the target encoder provides labels for training the context network, its gradients are blocked through its branch to prevent representation collapse, and its weights are updated using an EMA of the context encoder's parameters in (14). This

iterative process gradually enhances the context encoder's ability to produce meaningful low-dimensional embeddings from which the predictor infers future embeddings. Once trained, the proposed TS-JEPA maps high-dimensional device states to low-dimensional embeddings, enabling the autoregressive prediction of future embeddings by conditioning on predicted control commands.

In a nutshell, the proposed TS-JEPA for remote monitoring in wireless-networked control systems operates through two distinct phases.

- 1) *Training Phase*: During this phase, the devices transmit their high-dimensional states to remote controllers for control-command computation. Meanwhile, the proposed TS-JEPA is trained at the base station using a collected dataset, denoted as $\mathcal{D}_s = \{\mathbf{x}_{i,k}, \mathbf{u}_{i,k}\}_{k=1}^{K_s}$, where K_s represents the number of consecutive states and their corresponding control commands. The training process, detailed in Algorithm 1, iteratively optimizes the context encoder and predictor models until convergence, ensuring that the learned low-dimensional embeddings effectively minimize prediction errors.

Algorithm 1 Time-Series JEPA Algorithm

Require: High-dimensional state sequence $\{\mathbf{x}_{i,k}\}_{k=1}^{K_s}$, predicted control commands $\{\tilde{\mathbf{u}}_{i,k}\}_{k=1}^{K_s}$

Ensure: Trained parameters θ and φ

- 1: **Initialize:** Context encoder Ψ_θ , predictor $\mathcal{P}_\varphi$ with random weights
 - 2: Set target encoder $\Psi_{\bar{\theta}} \leftarrow \Psi_\theta$
 - 3: Set EMA decay rate $\eta \in [0, 1]$
 - 4: **for** each training step $k = 1$ to K_s **do**
 - 5: **Context Encoding:** $\mathbf{z}_{i,k} \leftarrow \Psi_\theta(\mathbf{x}_{i,k})$
 - 6: **Target Encoding:**
 - 7: **for** $j = 1$ to K_p **do**
 - 8: $\mathbf{z}_{i,k+j} \leftarrow \Psi_{\bar{\theta}}(\mathbf{x}_{i,k+j})$
 - 9: **end for**
 - 10: **Prediction:**
 - 11: Predict future embeddings as in eq. (12).
 - 12: **Loss cosine similarity computation**
 - 13: **Gradient Update:** Update θ and φ using gradient descent as in eq. (13)
 - 14: **EMA Update of Target Encoder** as in eq. (14)
 - 15: **end for**
 - 16: **Return** Trained Ψ_θ and $\mathcal{P}_\varphi$
-

- 2) *Inference Phase*: Once training is complete, the learned context encoder is deployed on the devices to encode high-dimensional states into low-dimensional embeddings. Meanwhile, the learned predictor is deployed at the cloud server to predict future embeddings by conditioning current embeddings on predicted control commands. This predictive capability enables the remote controller to predict latent dynamics, reducing reliance on frequent state transmissions and improving communication efficiency.

While these low-dimensional embeddings, whether received from the context encoder or predicted using the predictor,

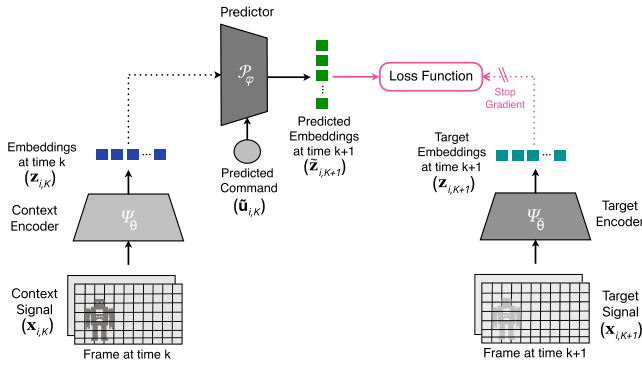


Fig. 2. TS-JEPA to encode high-dimensional state into low-dimensional semantic embeddings and predict the future semantic embeddings.

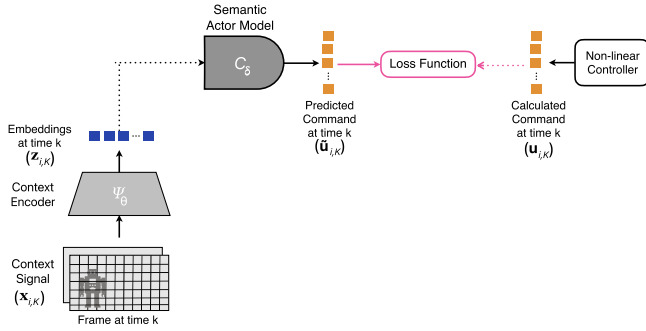


Fig. 3. Semantic actor model to predict control commands from low-dimensional semantic embeddings.

efficiently reduce wireless overhead, directly computing control commands from these embeddings poses a challenge. To address this challenge, a *semantic actor model* is introduced to map the low-dimensional embeddings to calculated control commands. This model effectively bridges the gap between the learned low-dimensional embeddings and control-command computation.

B. Semantic Actor Model

Once the TS-JEPA is well trained, the context encoder encodes high-dimensional states into low-dimensional embeddings. These embeddings are used to form a dataset $\mathcal{D}_a = \{z_{i,k}, u_{i,k}\}_{k=1}^{K_a}$, which contains K_a consecutive low-dimensional embeddings and their corresponding control command. To predict control commands directly from these low-dimensional embeddings, the *semantic actor model* $\mathcal{C}_\epsilon$, illustrated in Fig. 3, is trained using the dataset $\mathcal{D}_a$. The semantic actor model minimizes the average squared l_2 -distance between the predicted and target control commands, with its parameters optimized via gradient-based optimization as follows:

$$\arg \min_{\epsilon} \frac{1}{K_a} \sum_{k=1}^{K_a} \|u_{i,k} - \tilde{u}_{i,k}\|_2^2 \quad (15)$$

where $\tilde{u}_{i,k}$ is the predicted control command of the device i at time k .

In a nutshell, the remote control procedure employing the semantic actor model consists of two main phases.

- 1) *Training Phase*: During this phase, the devices transmit their high-dimensional states to remote controllers for control-command computation. Meanwhile, the proposed semantic actor model $\mathcal{C}_\epsilon$ is trained at the base station using the dataset $\mathcal{D}_a$ until convergence.
- 2) *Inference Phase*: Once the semantic actor model is well trained, it predicts control commands directly from low-dimensional embeddings. These embeddings may be received from the context encoder or predicted using the predictor in the proposed TS-JEPA. This predictive capability ensures continuous control under limited wireless resources or adverse channel conditions.

The proposed semantic-driven predictive control integrates TS-JEPA with a semantic actor model to ensure the robust control performance while minimizing communication overhead in large-scale wireless control deployments. Although the proposed framework efficiently leverages low-dimensional embeddings to improve communication efficiency, scalability, and control accuracy under limited wireless resources, optimizing wireless resource utilization remains crucial in large-scale deployments. To further improve communication efficiency, a channel-aware scheduling is introduced. This scheduling approach dynamically selects devices for transmission based on their channel conditions and AoI, ensuring that the most critical updates are prioritized for uplink transmission.

C. Channel-Aware Scheduling

The proposed channel-aware scheduling dynamically prioritizes devices for uplink transmission by jointly considering their channel conditions and AoI. The main objective is to minimize the long-term time-averaged transmission power while ensuring timely and reliable updates across the network. To achieve this, a centralized scheduler that determines both scheduling decisions and the corresponding transmission power in each time step to minimize the average transmission power under constraints that bound the long-term time-averaged AoI on each device below a predefined threshold, guarantee reliable transmission, and enforce feasible transmission power and scheduling decisions. This optimization problem, which ensures that the network resources are efficiently utilized while maintaining the information freshness and reliability, is given as

$$\text{Minimize } \limsup_{K \rightarrow \infty} \frac{1}{K} \sum_{k=1}^K \sum_{i=1}^I p_{i,k} \quad (16a)$$

$$\text{subject to: } \limsup_{K \rightarrow \infty} \frac{1}{K} \sum_{k=1}^K \beta_{i,k} \leq \beta_{i,\text{th}} \quad \forall i \quad (16b)$$

$$\gamma_{i,k} \geq \alpha_{i,k} \gamma_{\text{th}} \quad \forall i, k \quad (16c)$$

$$0 \leq p_{i,k} \leq \alpha_{i,k} p_{\text{max}} \quad \forall i, k \quad (16d)$$

$$\alpha_{i,k} \in \{0, 1\} \quad \forall i, k \quad (16e)$$

$$\sum_{i=1}^I \alpha_{i,k} \leq J \quad \forall k \quad (16f)$$

with variables $\{p_{i,k}, \alpha_{i,k}\}_{i \in \mathcal{I}}$. The constraint in (16b) limits the time-averaged AoI for each device below a predefined

threshold $\beta_{i,\text{th}}$ while the constraint in (16c) guarantees reliable transmission through ensuring that the received SNR for any scheduled device satisfies a threshold γ_{th} . Additionally, the constraints in (16d)–(16f) ensure the feasible scheduling variables to allow at most J devices to be scheduled at each time, and guarantee feasible transmission power, respectively. Note that the number of available resource blocks J is strictly less than the total number of devices I , reflecting the resource-constrained nature of the system. The AoI of the device i at time k , representing the time elapsed since the device's most recent update, quantifies the freshness of the received information given as follows [38]:

$$\beta_{i,k+1} = \begin{cases} 1 + \beta_{i,k}, & \text{if } \alpha_{i,k} = 0 \\ 1, & \text{otherwise.} \end{cases} \quad (17)$$

Since the optimization problem in (16) is a mixed integer nonconvex problem, we develop a dynamic control algorithm utilizing the Lyapunov drift-plus-penalty method [39]. According to the drift-plus-penalty method, the average AoI constraints are enforced by transforming them into queue stability constraints. For each inequality constraint in (16b), a virtual queue is associated in such a way that the stability of these virtual queues implies the feasibility of the average AoI constraint. For each device i , a virtual queue $Q_{i,k}$ is introduced and updated in each time step as follows:

$$Q_{i,k+1} = \max[Q_{i,k} - \beta_{i,\text{th}}, 0] + \beta_{i,k} \quad \forall i \in \mathcal{I}. \quad (18)$$

According to the mean-rate stability, the virtual queue is strongly stable if its average backlog is finite [39]. Note that the strong stability of the virtual queues in (18) implies that the average AoI constraint (16b) is satisfied. Next, we introduce the Lyapunov function and its drift, which are needed to define the queue stability.

Let $\mathcal{S}_k = \{\beta_{i,k}, \mathbf{H}_{i,k}\}_{i \in \mathcal{I}}$ be the network state at time k , and $\mathbf{Q}_k = [Q_{1,k}, Q_{2,k}, \dots, Q_{I,k}]$ be a vector containing all virtual queues. Then, a quadratic Lyapunov function $\mathcal{L}(\mathbf{Q}_k)$ is defined as follows:

$$\mathcal{L}(\mathbf{Q}_k) = \frac{1}{2} \sum_{i=1}^I Q_{i,k}^2. \quad (19)$$

The Lyapunov function measures the network congestion, where the Lyapunov function is small if all the queues are small and the Lyapunov function is large if at least one queue is large. Hence, by minimizing the expected change of the Lyapunov function from one step to the next step, the virtual queues $\{Q_{i,k}\}_{i \in \mathcal{I}}$ can be stabilized. The conditional Lyapunov drift $\Delta(\mathbf{Q}_k)$ is defined as the expected change in the Lyapunov function over one step given the current network state $\mathcal{S}_k$. Thus, $\Delta(\mathbf{Q}_k)$ is given as follows:

$$\Delta(\mathbf{Q}_k) = \mathbb{E}[\mathcal{L}(\mathbf{Q}_{k+1}) - \mathcal{L}(\mathbf{Q}_k) | \mathcal{S}_k]. \quad (20)$$

According to the drift-plus-penalty minimization, the dynamic control policy that minimizes the objective function of (16a) with constraints (16b)–(16f) is obtained by solving the optimization problem given as follows:

$$\text{Minimize } \Delta(\mathbf{Q}_k) + V \sum_{i=1}^I \mathbb{E}[p_{i,k} | \mathcal{S}_k] \quad (21a)$$

$$\text{subject to: (16c) —(16f)} \quad (21b)$$

with variables $\{p_{i,k}, \alpha_{i,k}\}_{i \in \mathcal{I}}$, where $V > 0$ denotes a control parameter that adjusts a tradeoff between the queue backlogs and objective function value.

Algorithm 2 Channel-Aware Scheduling Algorithm

```

1: Initialization:
2: Virtual queues  $Q_{i,0} = 0$  and AoI values  $\beta_{i,0} = 1, \forall i \in \mathcal{I}$ 
3: for each time step  $k = 1, 2, \dots$  do
4:   Observe channels  $\{H_{i,k}\}_{i \in \mathcal{I}}$  and AoI states  $\{\beta_{i,k}\}_{i \in \mathcal{I}}$ 
5:   for each device  $i = 1, \dots, I$  do
6:     Compute required power for reliability in (23)
7:     if  $p_{i,k}^{\text{req}} > p_{\text{max}}$  then
8:       Mark device  $i$  as infeasible for step  $k$ 
9:       Set  $S_{i,k} \leftarrow -\infty$ 
10:    else
11:      Compute drift-plus-penalty index in (25)
12:    end if
13:  end for
14:  Select up to  $J$  devices with the largest positive indices

```

$$S_{i,k} : \mathcal{B}_k = \text{Top-}J\text{-devices with } S_{i,k} > 0$$

```

15: for each device  $i \in \{1, \dots, I\}$  do
16:   if  $i \in \mathcal{B}_k$  then
17:     Set scheduling decision  $\alpha_{i,k} = 1$  and transmission
18:     power  $p_{i,k} = p_{i,k}^{\text{req}}$ 
19:     Update AoI:  $\beta_{i,k+1} = 1$ 
20:   else
21:     Set  $\alpha_{i,k} = 0$  and  $p_{i,k} = 0$ 
22:     Update AoI:  $\beta_{i,k+1} = \beta_{i,k} + 1$ 
23:   end if
24:   Update virtual queue in (18)
25: end for

```

Since minimizing the objective function in (21a) is intractable, we minimize an upper bound of (21a) given as follows:

$$(21a) \leq B + \mathbb{E} \left[\frac{1}{2} \sum_{i=1}^I \alpha_{i,k} \left(1 - (\beta_{i,k} + 1)^2 - 2Q_{i,k}\beta_{i,k} \right) + V \sum_{i=1}^I p_{i,k} \middle| \mathcal{S}_k \right]. \quad (22)$$

where the constant B in (22) is omitted since it does not affect the system performance in Lyapunov optimization. Having the upper bound in (22), instead of minimizing (16a), we minimize (22) subject to the constraints (16c)–(16f). Given the network state $\mathcal{S}_k$ at the beginning of each time step, we propose a low-complexity two-stage sequential optimization strategy to find a suboptimal solution to the joint power allocation and scheduling decision assignment problem. The method first assigns the minimum transmission power to meet the transmission reliability constraint in (16c) given as follows:

$$p_{i,k}^{\text{req}} = \frac{\gamma_{\text{th}} N_c}{|\mathbf{H}_{i,k}|^2} \cdot 10^{\frac{P_{\text{LOS}} - P_{\text{NLOS}}}{10}}. \quad (23)$$

If $p_{i,k}^{\text{req}} > p_{\max}$, the device is infeasible to transmit at this time step and cannot be scheduled. Given that the transmission power has been greedily allocated, the scheduling decision per-step combinatorial problem is given as follows:

$$\text{Minimize } \frac{1}{2} \sum_{i=1}^I \alpha_{i,k} \left(1 - (\beta_{i,k} + 1)^2 - 2Q_{i,k}\beta_{i,k} + V p_{i,k}^{\text{req}} \right) \quad (24a)$$

$$\text{subject to: (16d)–(16f)} \quad (24b)$$

with variable $\{\alpha_{i,k}\}_{i \in \mathcal{I}}$. The optimization problem in (24) can be solved by defining the drift-plus-penalty index as follows:

$$S_{i,k} = \left(1 - (\beta_{i,k} + 1)^2 - 2Q_{i,k}\beta_{i,k} + V p_{i,k}^{\text{req}} \right) \quad (25)$$

and set $S_{i,k} = -\infty$ if $p_{i,k}^{\text{req}} > p_{\max}$. In each time step k , the channel-aware scheduler schedules up to J devices with the largest $S_{i,k}$. Then, for every scheduled device, set the transmission power as $p_{i,k} = p_{i,k}^{\text{req}}$ with $p_{i,k}^{\text{req}} \leq p_{\max}$ to ensure transmission allocation feasibility in (16d). The steps of the channel-aware scheduling policy are summarized in Algorithm 2. By jointly considering transmission power and AoI, the proposed channel-aware scheduling effectively balances communication efficiency and control performance. This approach dynamically allocates resources to devices that require urgent updates, while ensuring reliable transmission. Hence, the proposed scheduling mitigates the risk of outdated control information and maximizes network resource utilization in large-scale deployments.

IV. SIMULATION RESULTS

To evaluate the performance of the proposed semantic-driven predictive control with channel-aware scheduling, we conducted extensive simulations on inverted cart-pole systems in a large-scale deployment. The inverted cart-pole system is adopted as a simulation benchmark due to its inherent instability and nonlinearity that capture the core challenges of many real-world industrial control systems. Its dynamics reflect the essential challenges of practical industrial systems, such as robotics manipulators and autonomous vehicles, where maintaining stability depends on timely and reliable communication between sensors, controllers, and actuators. Owing to these characteristics, the inverted cart-pole has been widely used in the literature to evaluate wireless-networked control systems under limited wireless resources [40], [41], [42]. Thus, it provides a representative testbed for validating the proposed framework's ability to balance communication efficiency and control performance under limited network capacity. In our simulation setup, each inverted cart-pole system's state is represented by an RGB frame captured at a fixed sampling interval of $\tau_o = 1$ ms over a time interval of 100 time steps. This sampling rate is selected to ensure capturing meaningful temporal dynamics for effective predictive embeddings learning, as detailed later. The corresponding control command applies a horizontal force to the cart to maintain control performance. The control commands are generated based on a nonlinear control policy with predefined control limits of $u_{\max} = 20$ N and $u_{\min} = -20$ N [43], [44].

TABLE II
TS-JEPA HYPERPARAMETERS

Hyperparameters	Values
Learning rate	2×10^{-1}
Batch size	256
Epochs number	150
Optimizer	SGD
Weight decay	0.0004
EMA decay rate	0.99

TABLE III
SEMANTIC ACTOR HYPERPARAMETERS

Hyperparameters	Values
Learning rate	6×10^{-3}
Batch size	200
Epochs number	300
Optimizer	AdamW
Dropout	0.2

A. Data Generation and Training

To train the proposed semantic-driven predictive control, we generated two distinct datasets: one for the TS-JEPA and another for the semantic actor model. The TS-JEPA dataset $\mathcal{D}_s$ includes 200 training and 40 testing trajectories, each consisting of RGB frames paired with their corresponding control commands. Meanwhile, the semantic actor model's dataset $\mathcal{D}_a$ comprises 100 training and 20 testing trajectories, each containing pairs of low-dimensional embeddings and their corresponding control commands.

The weight parameters of the proposed TS-JEPA are trained by minimizing the cosine similarity loss function defined in (13). The proposed TS-JEPA is trained using the hyperparameters listed in Table II, with a learning rate that decays by a factor of 0.99 every 20 epochs. Similarly, the semantic actor model is trained by minimizing the mean squared error (MSE) loss function defined in (15), using the hyperparameters in Table III. To mitigate overfitting, we apply early stopping based on validation performance. Each experiment is repeated five times to enhance statistical reliability, and the best results are reported.

The semantic-driven predictive control was implemented and trained on a NVIDIA Tesla V100-PCIE-16GB GPU-accelerated platform to handle computational complexity efficiently. The TS-JEPA encoder adopts a deep convolutional residual network (ResNet) architecture with layers of 64, 128, and 256 neurons, each followed by batch normalization and rectifier linear unit (ReLU) activation. The TS-JEPA predictor is structured as a multilayer perceptron (MLP) with a hidden layer of 1024 neurons and an output layer of 256 neurons. The semantic actor model is implemented as an MLP with two hidden layers containing 1024 and 256 neurons, respectively, each followed by an ReLU activation function.

To improve the robustness of the proposed TS-JEPA by enhancing spatial diversity, the RGB frames follow a series of pre-processing steps of image augmentation inspired by self-supervised learning approaches [15], [16].

TABLE IV
SYSTEM PARAMETERS

Parameters	Symbol	Values
Hall size		$300 \times 150 \text{ m}^2$
Room height		6 m
BS height	h_{BS}	10.0 m
Device height	$h_{i,R}$	1.5 m
Carrier frequency	W_c	3.75 GHz
Total bandwidth		20 MHz
Clutter height	h_c	3m
Clutter size	$D_{clutter}$	2.0 m
Clutter density	δ	60%
2D distance	D_i^{2D}	50 m
Noise power	N_c	-95 dB

- 1) *Color Jittering*: Randomly adjust brightness (0.05), contrast (0.1), saturation (0.1), and hue (0.05) across all pixels, applied in random order for each patch to enhance visual diversity.
- 2) *Color Dropping*: Converts frames to grayscale with a 0.05 probability, replacing RGB frame intensity with the luma component.
- 3) *Normalization*: Each color channel is normalized by subtracting the mean values [0.485, 0.456, 0.406] and dividing by the standard deviations [0.229, 0.224, 0.225] to improve the convergence.
- 4) *Resizing*: Frames are resized to 64×128 using a 5×5 Gaussian kernel with a standard deviation randomly sampled from [0.1, 0.2] to reduce computational overhead while keeping key visual features.

Additionally, control commands are pre-processed using z-score normalization to ensure stable TS-JEPA training. During TS-JEPA testing, resizing and normalization are applied to the RGB frames to ensure consistency with the training phase.

To evaluate the performance of the proposed framework under wireless conditions, simulations are conducted under varying target SNR values: $\gamma_{th} \in \{5, 10, 20\}$ dB. The remaining wireless parameters used in the simulation are detailed in Table IV.

B. Evaluation Metrics

To evaluate the performance of the proposed semantic-driven predictive control with channel-aware scheduling, five key metrics are employed: encoder performance, temporal consistency, prediction accuracy, control performance, and communication efficiency.

- 1) *Encoder Performance*: The quality of the trained context encoder of the proposed TS-JEPA is evaluated using t-Distributed Stochastic Neighbor Embedding (t-SNE). As a powerful dimensionality reduction technique, t-SNE maps high-dimensional embeddings into a 2-D space while preserving their local structure. This technique effectively reflects the encoder's ability to generate meaningful embeddings, where similar embeddings in the original high-dimensional space remain closely clustered in the reduced space. Improved clustering indicates

stronger semantic embeddings and improved encoder performance.

- 2) *Temporal and Spatial Consistency*: It is evaluated to quantify the relative change in state values across successive time steps. This metric is computed using the mean absolute percentage error (MAPE) between consecutive frames, defined as follows:

$$\mathcal{N}_{i,k}^p = \frac{1}{p} \sum_{v=1}^p \left| \frac{\mathbf{x}_{i,k}(v) - \mathbf{x}_{i,k-1}(v)}{\mathbf{x}_{i,k-1}(v)} \right| \times 100\% \quad (26)$$

where $\mathcal{N}_{i,k}^p$ represents the MAPE of device i at time k .

- 3) *Prediction Accuracy*: The accuracy of the proposed semantic actor model in predicting control commands is evaluated using the normalized mean absolute error (NMAE) between the predicted and ground truth control commands, defined as follows:

$$\mathcal{N}_{i,K_p}^u = \frac{\frac{1}{K_p} \sum_{k=K_s+1}^{K_s+K_p} |\tilde{u}_{i,k} - u_{i,k}|}{|\max(u) - \min(u)|} \quad (27)$$

where $\mathcal{N}_{i,K_p}^u$ quantifies the normalized prediction error of the i th device over the K_p prediction steps during testing. The normalization term ensures that the metric is independent of the control-command range, providing a fair evaluation across varying system scales.

- 4) *Control Performance*: The control performance is evaluated using a scoring function that rewards successful control outcomes based on the system's ability to drive the cart's position and pendulum angle to the desired position and angle. The scoring function is defined as follows:

$$\mathcal{R}_{i,k} = \begin{cases} 1, & |x_{i,k} - x_d| \leq 0.05 \ \& \ |\theta_{i,k}| \leq 0.05 \\ 0, & \text{otherwise} \end{cases} \quad (28)$$

where $\mathcal{R}_{i,k}$ represents the scoring function of the i th device at time k , $x_{i,k}$ is the cart's location, x_d is the desired cart position, and $\theta_{i,k}$ is the pendulum's angle.

- 5) *Communication Efficiency*: It is measured by the number of communication bits required for each device to transmit its state or corresponding embeddings at each step.

C. Baseline Models

To evaluate the performance of the proposed semantic-driven predictive control with channel-aware scheduling, we compare it with three control baselines and two scheduling baseline policies.

- 1) *Control Baselines*: The proposed semantic-driven predictive control is evaluated against the following control baselines.

- 1) *Baseline 1 (Optimal Control Policy)*: In this policy, the remote controller receives the high-dimensional state from the device and computes the corresponding control commands using a nonlinear control policy. This policy optimally balances state deviation and control effort, ensuring robust control performance. The calculated control command is then transmitted to the actuator

for application. This policy serves as a benchmark for optimal control performance under ideal conditions.

- 2) *Baseline 2 (Supervised Learning Model)*: This approach utilizes a supervised learning model to predict control commands directly from the high-dimensional state. The remote controller processes the received device state through the trained supervised model, which maps the device states to control commands. The predicted control command is then applied by the actuator.

- 3) *Baseline 3 (Generative Autoencoder Model)*: The generative autoencoder model employs an encoder–decoder structure. The encoder, deployed on the device, encodes the high-dimensional state into low-dimensional representations before transmitting them to the remote controller. The remote controller then decodes the received representations to reconstruct the device state, which is subsequently used to compute the control commands.

2) *Scheduling Baselines*: The proposed channel-aware scheduling is evaluated against the following scheduling baseline approaches.

- 1) *Baseline 4 (Round-Robin Scheduling)*: In this approach, each device periodically transmits its high-dimensional state to the remote controller following a predefined repeating order. The remote controller computes the control command, which is transmitted to the actuator for application. When a device is unscheduled, its actuator applies the most recently received control command [8], [9].

- 2) *Baseline 5 (Opportunistic Scheduling)*: This scheduling approach exploits channel conditions to determine when a device should transmit its state. When a device's channel conditions are poor, it remains unscheduled, and its actuator applies the previously received control command. By leveraging channel variations, this approach aims to improve transmission reliability and reduce communication overhead. However, frequent state updates may be missed if channel conditions remain unfavorable [45], [46].

The combination of these baselines provides a comprehensive evaluation of the proposed framework. The control baselines test the effectiveness of the semantic-driven predictive control, while the scheduling baselines evaluate the ability of the channel-aware scheduler to prioritize critical updates under limited network capacity.

D. Performance Evaluation

1) *Temporal and Spatial Consistency Analysis*: Fig. 4 illustrates the cdf of the MAPE between consecutive frames in the training dataset, evaluated at different sampling rates with and without image augmentation. As shown in Fig. 4(a), when no image augmentation is applied, decreasing the sampling rate leads to a lower temporal correlation between consecutive frames. This indicates that lower sampling rates promote greater temporal variation between frames, which is desirable for training the proposed TS-JEPa, as it encourages the learning of meaningful temporal dynamics rather than repetitive

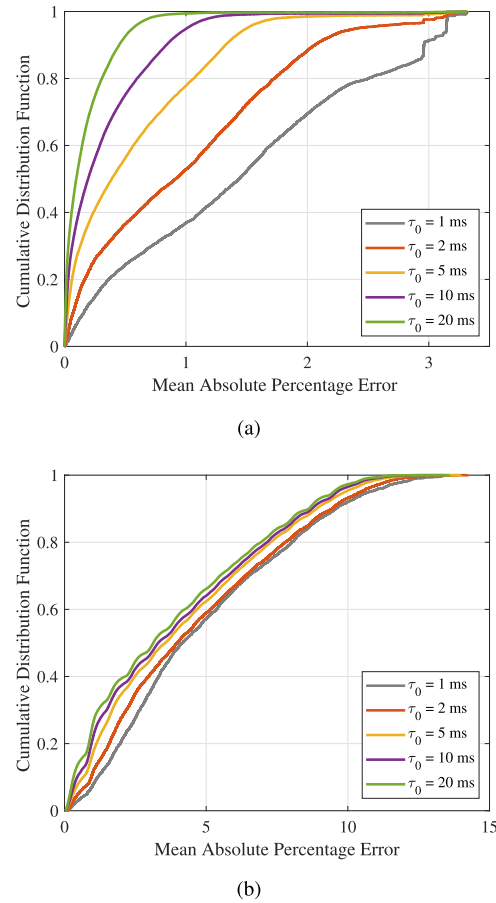


Fig. 4. CDF of the MAPE between consecutive frames in the training dataset under varying sampling rates. (a) Without image augmentation. (b) With image augmentation.

transitions. If the sampling rate is too high, consecutive frames become nearly identical, making it easier for the predictor to learn identity mappings, limiting the encoder's ability to extract semantically rich temporal representations.

Moreover, the application of image augmentation significantly increases the range of MAPE values, as shown in Fig. 4(b), with up to a fivefold increase compared to the one without image augmentation. This reflects improved spatial diversity across frames, which is critical for learning robust and generalizable embeddings. Therefore, the combination of a well-chosen sampling rate and image augmentation ensures that the frames fed into TS-JEPa are both temporally and spatially diverse. This diversity is key to effectively capture semantic representations and latent system behaviors within the embedding space, thereby improving the robustness and generalization capability of the proposed TS-JEPa across varying control environments.

2) *Encoder Performance*: Fig. 5 presents the t-SNE visualization of learned embeddings from twenty trajectories, comparing the context encoder of the proposed TS-JEPa with the encoder of a generative autoencoder model. The embeddings produced by the proposed TS-JEPa exhibit distinct and well-clustered structures, indicating that it effectively captures meaningful semantic embeddings of the device states. In contrast, the embeddings generated by the generative autoencoder model exhibit weaker clustering and greater overlap, indicating

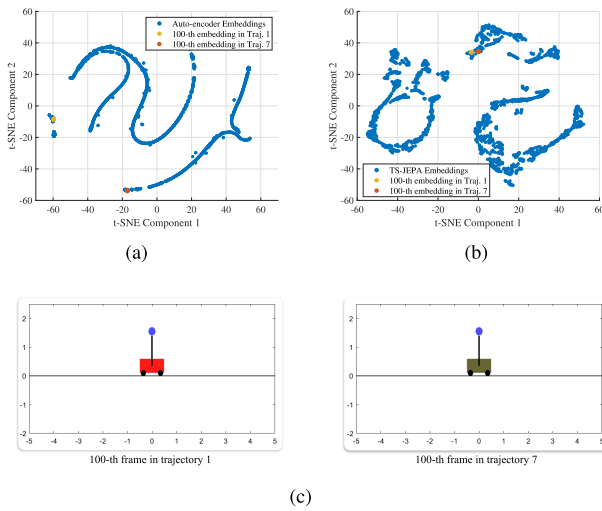


Fig. 5. t-SNE visualization of learned embeddings from 20 trajectories. (a) Generative autoencoder model. (b) Proposed TS-JEPA model. (c) Example of two distinct frames in different trajectories whose embeddings appear in (a) and (b).

a limited ability to distinguish between semantically different device states within embedding space.

To further evaluate the semantic encoding capability of the proposed TS-JEPA, we highlight embeddings corresponding to two different frames (with varying cart colors) that share identical device states (cart location and pendulum angle), as shown in Fig. 5(c). The proposed TS-JEPA maps these frames to nearby points in the embedding space, indicating that it captures the underlying semantic embeddings within the frames irrespective of visual differences. Conversely, the generative autoencoder model maps them to distant embeddings, revealing its sensitivity to superficial variations. These results demonstrate the robustness of the proposed TS-JEPA in learning semantically invariant embeddings, which is crucial for control tasks. By directly leveraging these low-dimensional embeddings for control-command prediction, the proposed TS-JEPA avoids the need to reconstruct high-dimensional states, as required by generative models, achieving computational efficiency and improved performance in downstream control applications.

3) *Encoding and Prediction Capability Evaluation*: Fig. 6 compares the proposed semantic-driven predictive control with several control baseline models in a single-device scenario, evaluating command prediction accuracy, communication cost, and normalized control score across the prediction horizon. To validate the encoding capability of the proposed approach, we first consider a no-prediction case where the remote controller receives either high-dimensional states or low-dimensional embeddings at every time slot. The proposed TS-JEPA with two consecutive frames ($\kappa = 2$) is compared to: 1) a supervised learning model with $\kappa = 2$ and $\kappa = 4$; 2) a generative autoencoder with $\kappa = 2$; and 3) the optimal nonlinear control policy. As shown in Fig. 6(a), the proposed TS-JEPA achieves the lowest normalized control prediction error, closely matching the optimal policy while requiring significantly fewer communication bits [see Fig. 6(b)]. This leads to a normalized control score [see Fig. 6(c)] that is nearly

optimal, validating the TS-JEPA encoder's ability to extract semantic embeddings from high-dimensional frames and support efficient remote control via the semantic actor model.

In contrast, the generative autoencoder model exhibits higher control prediction error due to reconstruction inaccuracies and limited temporal data, although its score surpasses that of the supervised learning baseline. This is attributed to its use of a nonlinear control policy during control-command computation. Meanwhile, the supervised model suffers from poor generalization due to limited training diversity, requiring more high-dimensional frames for improved prediction. Increasing κ from 2 to 4 enhances control performance but incurs higher communication costs, emphasizing the tradeoff between communication overhead and control performance in generative approaches.

To evaluate the prediction capability of the proposed TS-JEPA, we evaluate a case where the device transmits only at the initial time slot, and the remote controller needs to predict future control commands. In this case, the proposed TS-JEPA is compared to the control baseline models with repetitive-action and zero-action strategies. The results show that the proposed TS-JEPA maintains the lowest control prediction error throughout the prediction horizon, outperforming other control baseline models whose errors grow due to the lack of dynamic inference. This is enabled by TS-JEPA's ability to predict future embeddings using its predictor and compute control commands using the semantic actor model. Although TS-JEPA's prediction error increases slightly over time due to autoregressive accumulation, it achieves better normalized scores than all baselines, including the optimal policy. This is due to its aggressive control behavior, which rapidly drives the device toward the desired state. Moreover, the proposed TS-JEPA achieves these results with significantly fewer communication bits, demonstrating its effectiveness in reducing communication overhead without compromising the control performance. These results underscore the strength of the proposed TS-JEPA in capturing and predicting semantic embeddings, enabling scalable and communication-efficient control, particularly in bandwidth-constrained wireless control systems.

4) *Embedding Dimension Evaluation*: Fig. 7 investigates the impact of the embedding dimension on the performance of the proposed semantic-driven predictive control in a single-device scenario, evaluating command prediction accuracy, communication cost, and control performance across the prediction horizon. To determine the appropriate embedding dimensionality, a grid-search strategy is employed over a small set of candidate dimensions. The proposed TS-JEPA is trained under identical configurations for each candidate dimension, and the embedding dimension yielding the Pareto-optimal balance, i.e., the lowest control prediction error and the highest normalized control score across the prediction horizon, is selected for deployment. As shown in Fig. 7(a), increasing the dimensionality of the embedding significantly reduces the normalized control prediction error, highlighting the importance of higher dimensional embeddings in capturing richer semantic features from high-dimensional frames, which improves the accuracy of downstream control-command prediction. However, as shown in Fig. 7(b), this improvement

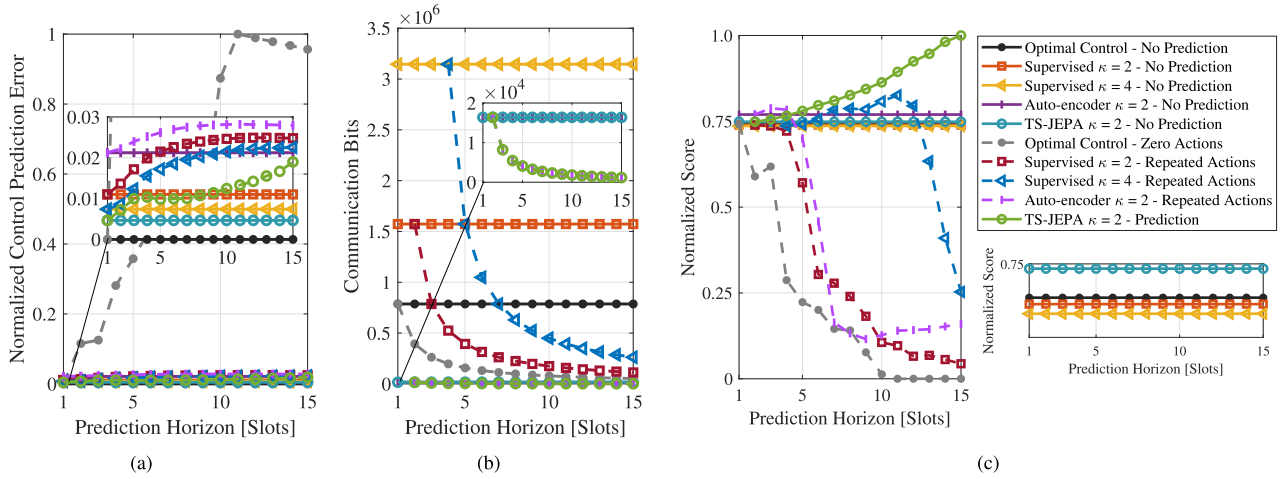


Fig. 6. Comparison between the proposed semantic-driven predictive control and different control baselines in a special case of a single device in terms of (a) command prediction accuracy, (b) communication cost, and (c) control accuracy.

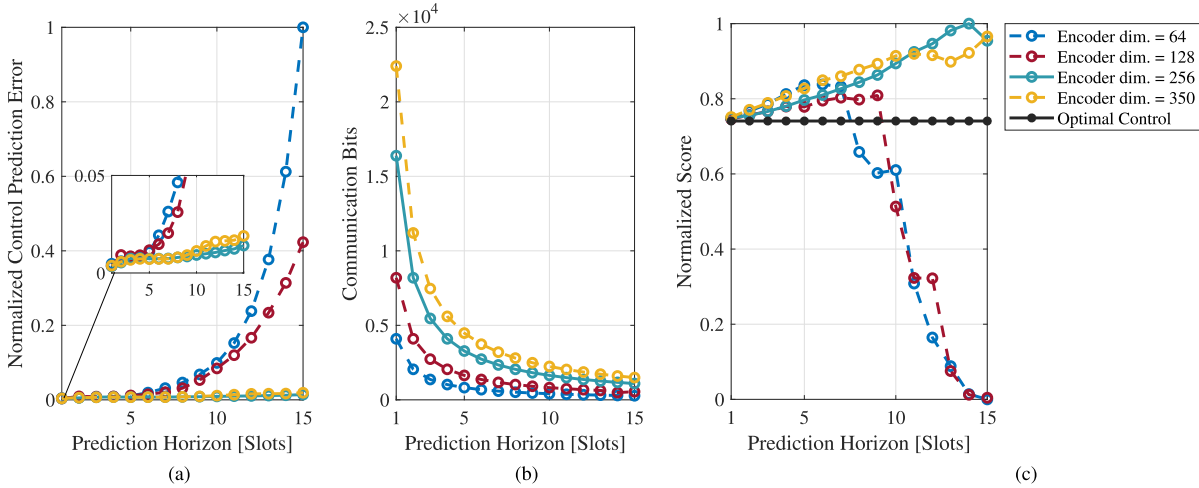


Fig. 7. Embedding dimensions impact for a single device in terms of (a) command prediction accuracy, (b) communication cost, and (c) control cost.

in semantic representations comes at the cost of increased communication overhead. Larger semantic embeddings require more communication bits to transmit, which is a critical consideration under limited network capacity. Nevertheless, Fig. 7(c) demonstrates that higher dimensional embeddings also prolong the normalized control score over the prediction horizon, indicating a more stable control performance. These results highlight a fundamental tradeoff between embedding dimensionality, control accuracy, and communication efficiency. Hence, careful selection of the embedding dimension via the proposed grid-search procedure is essential to ensure the semantic-driven predictive control remains both scalable and resource-efficient.

5) *Training Dataset Size Evaluation:* Fig. 8 evaluates the impact of training dataset size on the performance of the proposed semantic-driven predictive control in a single-device scenario, focusing on command prediction accuracy, communication cost, and control performance over the prediction horizon. As illustrated in Fig. 8(a), increasing the training dataset size significantly improves the normalized control prediction accuracy. This improvement indicates that a sufficiently large dataset is essential to capture the underlying latent

dynamics critical to downstream control-command prediction. However, as shown in Fig. 8(b), this enhanced accuracy comes at the cost of a higher communication overhead. Larger training datasets typically lead to richer semantic embeddings, which may require more bits to transmit, posing a challenge in resource-constrained wireless control systems. In contrast, training with insufficient data tends to underfit, leading to noisy semantic embeddings and poor generalization. In the context of embedding prediction, limited training data can also cause representation collapse, degrading semantic richness and control robustness. Despite the communication cost, Fig. 8(c) demonstrates that training with larger datasets maintains a higher normalized control score overextended prediction horizons, reflecting improved long-term control performance and predictive robustness. In contrast, smaller training datasets result in a rapid decline in the normalized control score, signaling deterioration in control performance under long-horizon prediction due to poor generalization. These results highlight a key tradeoff: while increasing the training dataset size improves control accuracy and robustness, it also increases communication overhead. Therefore, careful selection of the training dataset size is necessary to balance

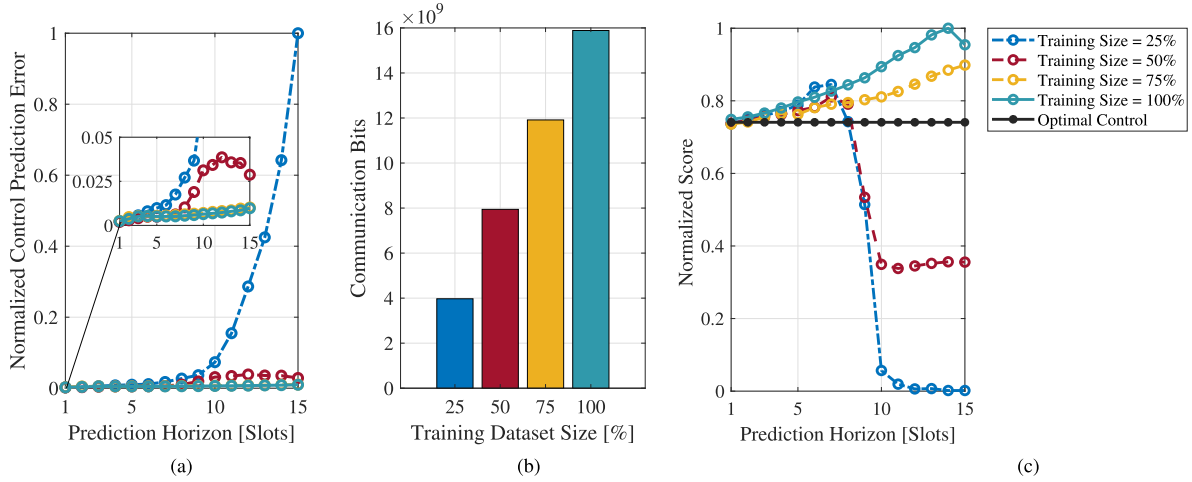


Fig. 8. Training dataset size impact for a single device in terms of (a) command prediction accuracy, (b) communication cost, and (c) control cost.

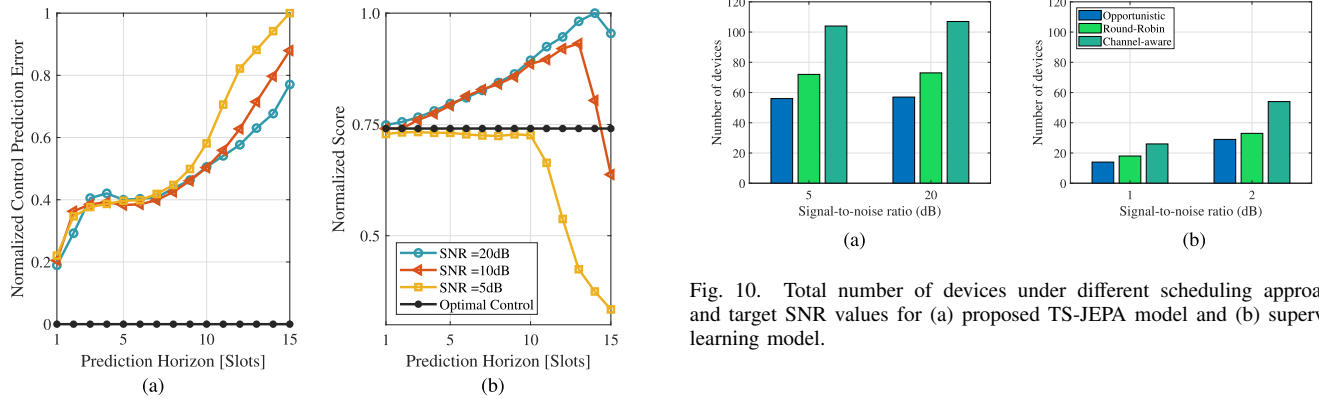


Fig. 9. Impact of target SNR in a single device in terms of (a) command prediction accuracy and (b) control cost.

between control performance and communication efficiency. Optimizing this tradeoff is essential for ensuring the scalability and practicality of the proposed semantic-driven predictive control in large-scale, bandwidth-constrained wireless control systems.

6) *Target SNR Evaluation*: Fig. 9 evaluates the impact of target SNR values on the performance of the proposed semantic-driven predictive control in a single-device scenario, focusing on command prediction accuracy and control performance over the prediction horizon. As illustrated in Fig. 9(a), lowering the target SNR from 20 to 5 dB significantly degrades the control prediction accuracy. This degradation arises because lower SNR values result in fewer successfully received samples during training, limiting the TS-JEPA model's ability to capture the underlying latent dynamics necessary for accurate control-command prediction. Correspondingly, Fig. 9(b) shows a decline in normalized control scores under low SNR values, further confirming the negative impact on long-horizon control performance. Fewer training samples impair the model's generalization capability, leading to reduced predictive accuracy and higher normalized control prediction error over time. In contrast, training the TS-JEPA under high SNR values captures the

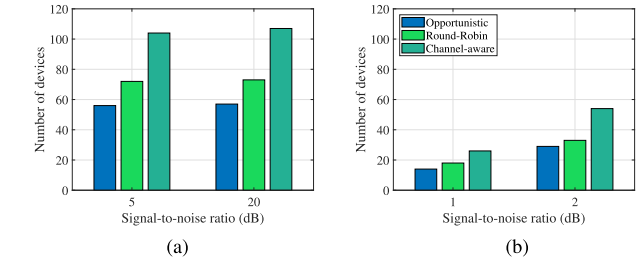


Fig. 10. Total number of devices under different scheduling approaches and target SNR values for (a) proposed TS-JEPA model and (b) supervised learning model.

underlying latent dynamics, enabling more robust and accurate control overextended horizons. This results in a higher normalized control score, reflecting robust prediction accuracy and effective control overextended prediction horizons. However, achieving high SNR value requires increased wireless resource consumption, highlighting a key tradeoff: improving predictive control robustness versus limiting communication cost. These findings emphasize the importance of considering SNR values during model training, particularly in bandwidth-constrained wireless control systems.

7) *System Scalability With Encoding Capability*: Fig. 10 presents the scalability performance of the proposed channel-aware scheduling compared to baseline scheduling approaches. The evaluation examines the maximum number of devices that can be supported while maintaining acceptable control performance, defined as a normalized score within the specified range [0.74, 1.0], across different SNR values for the proposed TS-JEPA and supervised learning models. The results demonstrate that the proposed scheduling, when combined with the TS-JEPA, significantly outperforms round-robin and opportunistic baselines in terms of scalability under different SNR values. This superior performance is attributed to two key design features. First, the TS-JEPA effectively compresses high-dimensional states into low-dimensional semantic embeddings, reducing transmission overhead and enabling support for a larger number of devices. Second, channel-aware scheduling dynamically selects devices for uplink transmission

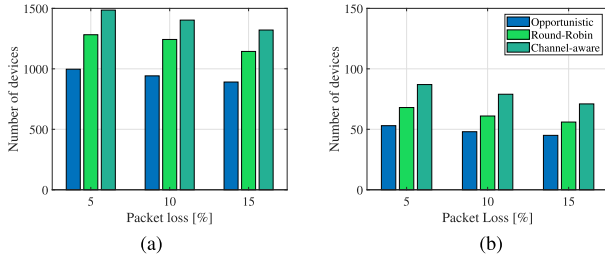


Fig. 11. Total number of devices under different scheduling approaches and packet losses for (a) proposed TS-JEPA model and (b) supervised learning model.

based on their AoI and channel conditions, ensuring timely and reliable updates. In contrast, round-robin scheduling offers moderate scalability due to its fairness in transmission updates, but suffers from inefficiencies caused by ignoring channel conditions and device urgency. The opportunistic scheduling, despite leveraging channel conditions, exhibits the lowest scalability. This is attributed to its tendency to favor devices with good channels, regardless of their update criticality, resulting in poor control performance and suboptimal resource utilization in large-scale deployment. Moreover, the comparison between the TS-JEPA and supervised learning models highlights the robustness of the proposed semantic-driven control under varying SNR values. The TS-JEPA model's semantic embeddings allow for efficient operation even under low SNR values, while the supervised baseline model struggles due to the need to transmit high-dimensional states, consuming more bandwidth and restricting the number of supported devices. These results underscore the effectiveness of combining semantic compression with channel-aware scheduling to provide scalable and reliable control in bandwidth-constrained wireless networks.

8) *System Scalability With Prediction Capability*: Fig. 11 evaluates the scalability of the proposed channel-aware scheduling against baseline approaches under varying packet losses. The evaluation focuses on determining the maximum number of devices that can be reliably supported while maintaining the acceptable control performance, defined by a normalized score within the specified range [0.74, 1.0], for both the proposed TS-JEPA and supervised learning models. The results demonstrate that the proposed framework, combining semantic-driven predictive control with channel-aware scheduling, substantially outperforms round-robin and opportunistic scheduling approaches with the supervised learning model in terms of scalability, even under adverse network conditions. This superior performance is attributed to two design features. First, TS-JEPA encodes high-dimensional states into low-dimensional semantic embeddings, significantly reducing communication overhead and enabling the support of more devices. Moreover, its predictive capability allows it to predict future embeddings even when packets are lost, thereby maintaining an acceptable control performance without requiring retransmission. Second, channel-aware scheduling dynamically prioritizes transmissions based on device-specific AoI and channel conditions, ensuring timely and reliable updates. In contrast, the supervised learning model exhibits limited robustness and scalability, primarily due to its dependence

on transmitting high-dimensional states, which burdens network capacity and lacks predictive capability under packet loss. These results highlight the effectiveness of integrating semantic-driven predictive control with channel-aware scheduling to enable robust and scalable control across large-scale wireless-networked systems operating under limited network capacity.

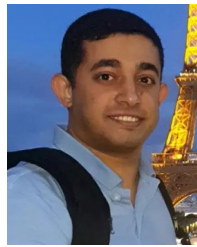
V. CONCLUSION

This work introduced a novel semantic-driven predictive control, integrated with channel-aware scheduling, to address the core challenges of control robustness, communication efficiency, and scalability in wireless-networked control systems. The proposed approach employs a self-supervised TS-JEPA with a semantic actor model to encode high-dimensional sensory data into low-dimensional semantic embeddings, significantly reducing uplink communication overhead. Beyond efficient encoding, TS-JEPA enables forward prediction of future embeddings, allowing the remote controller to infer future embeddings without continuous uplink updates. To further enhance communication efficiency, a channel-aware scheduling dynamically prioritizes embedding transmission devices based on channel quality and AoI, ensuring timely updates where they are most critical. Simulation results on large-scale inverted cart-pole systems demonstrate that the proposed framework not only achieves high control accuracy and low prediction error but also dramatically reduces communication costs compared to conventional baselines. Furthermore, the proposed framework supports robust control performance overextended prediction horizons and scales efficiently to accommodate significantly more devices in a limited uplink capacity. As future work, we plan to linearize the nonlinear TS-JEPA predictor to enable stability analysis within the latent space.

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